

CFD-DEM numerical investigation of the effects of water content and inclination angle on interactions between debris flows and slit dam

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ABSTRACT

Debris flows, composed of sediment–water mixtures, can pose significant risks to downstream populations and infrastructure in mountainous regions. Slit dams have been demonstrated as being effective measures for decreasing the destructivity of debris flows. Reliable evaluations of the interactions between debris flows and slit dams are challenging if the complicated component characteristics of debris flows, such as the water content, are not considered. Here, a multi-scale study of debris flows impacting a slit dam is conducted via computational fluid dynamics and discrete element method (CFD-DEM) simulations to understand the influences of the water content (controlled by solid volume fraction in this study) and the channel inclination angle on flow–slit dam interactions. The results indicate that both the initial solid volume fraction and the channel inclination angle have significant effects on the impact kinematics of flow–slit dam interactions and regulation functions of the slit dam: (i) For gentle slopes, the deposition of debris flows is primarily influenced by flume basal friction, and the normalized peak total impact forces have little difference for a given inclination angle, whereas for steep slopes, an overtopping can occur, further result in the increasing of the normalized peak total impact forces with the increase of solid volume fraction. (ii) The run-up height and retention efficiency are also analyzed to provide a scientific reference for optimizing the design of slit dams.

1. Introduction

Debris flows, which occur in mountainous regions, are rapid gravity-driven two-phase flows of sediment–water mixtures that are extremely destructive and pose significant risks to downstream populations and infrastructure (Chen et al., 2019; Iverson, 1997; Ng et al., 2016). Slit dams or slit-type barriers, typically constructed with steel or reinforced concrete, are widely employed as effective countermeasures for impeding and regulating debris flows in gullies (Bernard et al., 2019; Vagnon, 2020). These structures exhibit exceptional resistance to high dynamic impact forces while concurrently mitigating downstream peak flow rates and effectively capturing granular materials (Goodwin and Choi, 2020; Huang et al., 2022; Kong et al., 2022; Rossi and Armanini, 2019). However, the credible evaluation and assessment of flow–slit dam interactions still pose significant challenges due to the complex composition of debris flow, intricate multi-stage interaction process, numerous coupling influence factors, and complex multi-scale nature

(Goodwin and Choi, 2020; Ren et al., 2023).

Therefore, relevant studies have been undertaken to enhance our comprehension of the interactions between debris flows and slit dams or slit-type barriers. These efforts have involved field survey (Leonardi and Pirulli, 2020; Shima et al., 2016), theoretical analysis (Piton et al., 2022; Proske et al., 2011), experimental tests (Li et al., 2023; Lin et al., 2017; Ng et al., 2014; Zhou et al., 2019), and numerical simulations (Huang et al., 2023; Kong et al., 2022; Ren et al., 2023; Xiong et al., 2023) aimed at identifying key features and influencing factors. Table 1 lists the partial previous studies on slit dam or slit-type barriers based on their research methods. In general, although the existing literature about experiments extensively investigates the interaction issues between flow and slit dam, most of them primarily focus on examining the effects of flow properties (e.g., grain size distributions or shape of solid phase, fluid content, fluid viscosity, and Fr conditions) as well as structural configurations of slit dams or slit-type barriers (e.g., relative slit size, number of slits, and slit separation) on impact forces and regulating

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Table 1

Summary of partial previous studies on slit dam or slit-type barriers.

Study type	References
Theoretical analysis	Proske et al. (2011); Piton et al., (2022)
Field survey & monitoring	Shima et al., (2016); Leonardi and Pirulli (2020)
Flume tests	Choi et al. (2014); Ng et al. (2014); Silva et al., (2016); Lin et al., (2017); Sun et al., (2018); Rossi and Armanini, (2019); Zhou et al., (2019); Hu et al., (2020); Goodwin and Choi, (2021); Sun et al., (2021); Huang et al., (2022); Li et al., (2023)
Centrifuge test	Huang et al., (2023)
Depth average	Kattel et al., (2018)
SPH	Silva et al., (2016)
DEM	Ng et al. (2014); Choi et al. (2015); Choi et al. (2016); Leonardi et al., (2019); Goodwin and Choi, (2020); Marchelli et al., (2020); Zhou et al., (2020); Gong et al., (2021); Goodwin and Choi, (2021); Huang et al., (2022); Huang et al., (2023)
Coupling methods	CFD-DEM
	Kong et al., (2022); Kong and Guan, (2023); Mao et al., (2023)
	MPM-DEM
	Ren et al., (2023)
	SPH-DEM
LBM-DEM	Xiong et al., (2023)
	Marchelli and De Biagi (2019)

functions, such as mass outflow rate, and retention efficiency. Nevertheless, the study of flow-slit dam interaction issues through experiments presents several challenges. Investigating the interaction between the fluid and solid phases in debris flow, as well as understanding their respective influence mechanisms on the flow impact process, poses significant hurdles.

To achieve this objective, numerical simulation emerges as a promising option. However, the existing literature about numerical simulations predominantly focuses on dry granular flows, thereby limiting its applicability to natural debris flows that incorporate fluid dynamics (Choi et al., 2015, 2016; Gong et al., 2021; Goodwin and Choi, 2020; Huang et al., 2023; Zhou et al., 2020). For instance, Leonardi et al., (2019) conducted a parametric investigation on the load exerted by dry granular flows on slit dams, considering grain size, slit size, and channel slope. Marchelli et al., (2020) conducted an analysis on the interaction between dry granular flows and slit dams, and presented a design diagram illustrating the clogging mechanisms based on channel slope and slit sizes using DEM. In recent years, the integration of discrete models and continuum models in CFD-DEM facilitates a direct representation of

the inherent two-phase mixing characteristics observed in natural debris flows, and has been successfully used in this field. For example, Fang et al., (2021, 2022) analyzed the effects of solid fraction and fluid properties (viscosity and density) on the impact force on rigid barriers. Kong et al., (2022) presented the first CFD-DEM simulations on flow-dam interactions, encompassing slit, flexible, and rigid dams. However, their work primarily focuses on impact loads and run-up heights. Similarly, using the CFD-DEM method, Kong and Guan, (2023) compiled the first unified design diagrams that quantitatively establish links between flow properties, slit size, and key design indices such as over-spilling dynamics, downstream momentum reduction ratio, and retention efficiency. Mao et al., (2023) further conducted CFD-DEM simulation to explore the impact mechanism and load applied to a slit dam. Nevertheless, insufficient attention has been given to exploring the influence of channel inclination angle, thus leaving this aspect unclear (Leonardi et al., 2019; Lin et al., 2017; Marchelli et al., 2020). Therefore, this study employs a CFD-DEM model to comprehensively simulate the entire flow process of debris flows and investigate the influence of water contents and channel inclination angle on the intricate interaction between debris flows and slit dam.

The following is the organization of the paper. First, we describe the CFD-DEM framework used in this paper. Subsequently, the CFD-DEM coupling method is validated via a single sphere sinking in water. Then, the setup and the geometries of the flume model and the relevant parameters concerning the materials and simulations are presented. Subsequently, the influence of channel inclination angles and water content on the interaction behaviors of debris flows and slit dams is quantitatively analyzed. On the basis of the results, rational recommendations for slit dams are proposed to improve engineering design.

2. Methods

The coupling between CFD and DEM is realized via OpenFOAM (Weller et al., 1998) and LIGGGHTS (Kloss et al., 2012) that has been widely used (see Fig. 1). The former is a computational fluid dynamics (CFD) software based on finite volume method, while the latter is used for DEM. The data exchange between them is implemented by CFDEMcoupling (Goniva et al., 2012).

2.1. CFD-DEM coupling method

In CFD, the local-averaged Navier–Stokes equations with the volume of fluid (VOF) method are used:

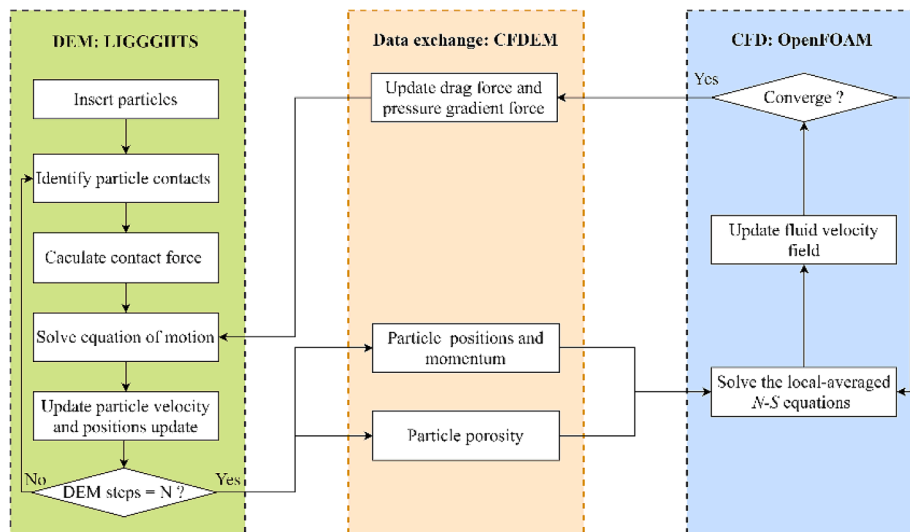


Fig. 1. Flowchart of the non-resolved computational fluid dynamics and discrete element method (CFD-DEM) coupling engine in CFDEMcoupling.

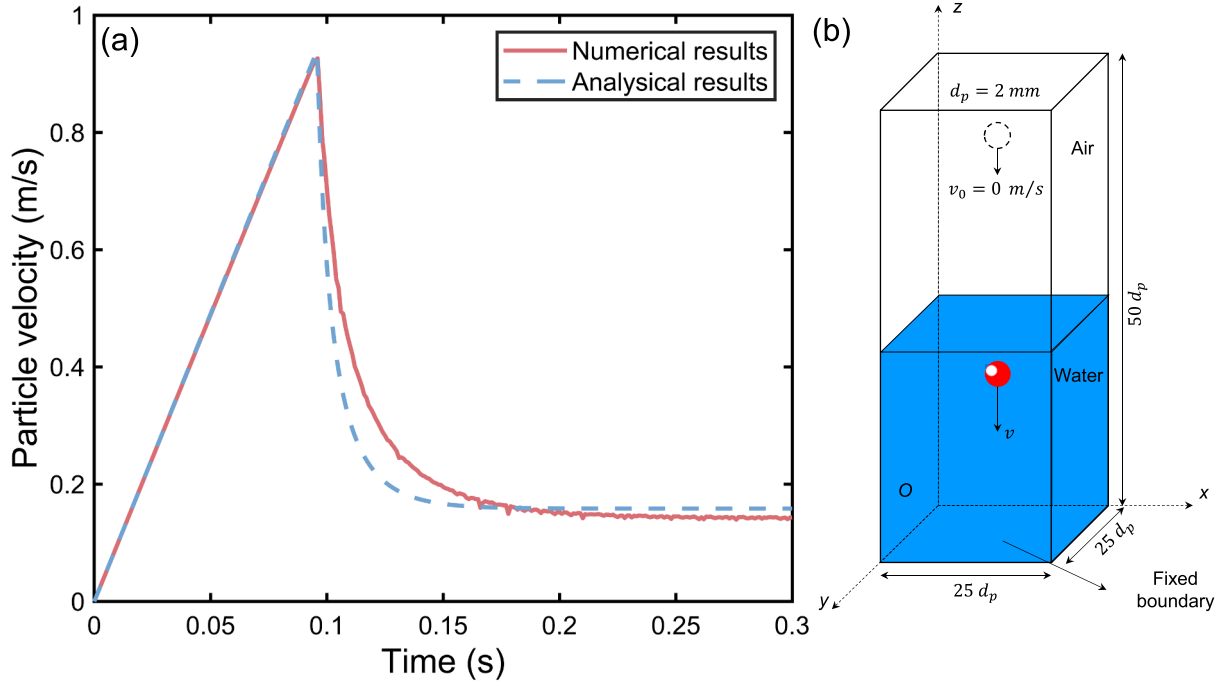


Fig. 2. (a) Comparison between the numerical and analytical results in terms of the particle settling velocity, and (b) model setup of a single spherical particle settling in a water tank.

$$\frac{\partial}{\partial t}(\alpha_f \rho_f \mathbf{u}_f) + \nabla \hat{\mathbf{A}} \cdot (\alpha_f \rho_f \mathbf{u}_f \mathbf{u}_f) = -\alpha_f \nabla p + \alpha_f \nabla \hat{\mathbf{A}} \cdot \mathbf{T}_f + \alpha_f \rho_f \mathbf{g} - \mathbf{f}_{pf}, \quad (1)$$

$$\frac{\partial \alpha_f}{\partial t} + \nabla \hat{\mathbf{A}} \cdot (\alpha_f \mathbf{u}_f) = 0, \quad (2)$$

where α_f is the volume fraction of water in each discrete cell, $\mathbf{u}_f$ is velocity of the water, ρ_f is the density, p is the water pressure, $\mathbf{T}_f$ is the stress tensor with viscosity μ , $\mathbf{f}_{pf}$ is the interaction force between particle and fluid, and $\mathbf{g}$ is the gravity.

The particle motion in DEM is governed by Newton's equations:

$$m_i \frac{d\mathbf{u}_i}{dt} = \sum_{j=1}^{n-1} \mathbf{F}_{ij}^p + \mathbf{F}_i^f \quad (3)$$

$$I_i \frac{d\boldsymbol{\omega}_i}{dt} = \sum_{j=1}^{n-1} \mathbf{M}_{ij}^p, \quad (4)$$

where m_i is the mass of the i -th particle; $\mathbf{u}_i$ and $\boldsymbol{\omega}_i$ are the translational velocities and rotational velocities, respectively; I is the moment of inertia; $\mathbf{F}_{ij}^p$ indicates the contact forces generated by particles or structure, calculated by the Hertz model; $\mathbf{F}_i^f$ is the fluid force, including drag force $\mathbf{F}_i^d$ and the buoyancy force $\mathbf{F}_i^b$ and $\mathbf{M}_{ij}^p$ indicates the torque.

A momentum exchange term in CFD is used to consider the fluid-particle interaction:

$$\mathbf{f}_{pf} = K_{pf}(\mathbf{u}_f - \langle \mathbf{u}_p \rangle), \quad (5)$$

where $\langle \mathbf{u}_p \rangle$ is the average particle velocity in a cell and K_{pf} is shows as

$$K_{pf} = \frac{\sum_{i=1}^l F_i^d}{V_{fc} |\mathbf{u}_f - \langle \mathbf{u}_p \rangle|}, \quad (6)$$

where l is the particle number in a discrete cell, V_{fc} is the cell volume, and F_i^d is the drag force of particles in DEM.

The Di Felice model is used to calculate drag force:

$$\mathbf{F}_i^d = \frac{1}{8} C_d \rho_f \pi d_i^2 |\mathbf{u}_f - \mathbf{u}_i| \alpha_f^{1-\chi}, \quad (7)$$

with

$$C_d = \left(0.63 + \frac{4.8}{\sqrt{Re_p}} \right)^2 \quad (8)$$

$$\chi = 3.7 - 0.65 \exp \left[- \frac{(1.5 - \log_{10} Re_p)^2}{2} \right] \quad (9)$$

where the particulate Reynolds number is

$$Re_p = \frac{\alpha_f \rho_f d_i |\mathbf{u}_f - \mathbf{u}_i|}{\mu_f} \quad (10)$$

where d_i is the particle diameter, C_d is the drag coefficient, and χ is a corrective coefficient.

In addition, the buoyant force of a single particle is calculated as follows

$$\mathbf{F}_i^b = \frac{1}{6} \pi d_i^3 \rho_f \mathbf{g} \quad (11)$$

Note that Kong et al., (2023) calculated four fluid–solid interaction forces, including drag, buoyant, and the viscous, virtual mass forces which are not considered in this study, for modeling flow-barrier interaction issues. And it is worth noting that the unresolved approach, which calculates the force exerted on particles by fluids using empirical drag force models, cannot accurately capture the complex flows around individual particles (Nguyen et al., 2021). Additionally, due to the coarser CFD mesh used in this method, there may be a lack of clarity at the liquid phase interface captured by VOF method, which could affect calculations of fluid-particle interactions near the liquid surface. Therefore, we adopted a computational cell size of $3d_s$ for our CFD simulations in line with many other unresolved CFD-DEM studies

Table 2

Input parameters of the benchmark simulation.

DEM parameter	Value	CFD parameter	Value
Particle diameter d_s (mm)	2	Fluid density ρ_f (kg/m ³)	1000
Particle density ρ_s (kg/m ³)	2650	Viscosity μ_f (Pa·s)	0.001
Young's modulus E_s (MPa)	5	Simulation parameters	Value
Poisson's ratio ν	0.45	Gravity g (m/s ²)	9.81
Coefficient of friction μ_s	0.5	DEM time step size Δt_s (s)	10 ⁻⁶
Coefficient of restitution e	0.3	CFD time step size Δt_f (s)	10 ⁻⁴

(Shen et al., 2022).

2.2. Coupling method validation

To test the validity of the CFD-DEM, a benchmark case of a single particle settling into water was performed. In this case, a sphere with diameter of $d_s = 2$ mm was dropped from a position 5 mm under the top surface of a water container (see inset of Fig. 2) having a dimension of 50 mm $\times$ 50 mm $\times$ 100 mm. The initial velocity was set to zero. Table 2 lists the input parameters used in the simulation. During the settling of the particle, the settling velocity was recorded. Fig. 2 shows a comparison between the numerical results and the analytical results about the particle settling velocity. The strong correlation between the numerical and analytical results serves as proof of the reliability of the CFD-DEM method for problems with moving particles in a fluid.

2.3. Model setup and parameters

Fig. 3 depicts the configuration of the CFD-DEM model employed for simulating the impact of debris flows on a slit dam. The flume model has dimensions of 5.0 m $\times$ 0.3 m $\times$ 1.0 m ($L_f \times W_f \times H_f$). The channel width, which is designed to be 30 times the grain diameter d_s , is chosen with the intention of minimizing the impact of the lateral walls on the domain. Upstream, a release gate is positioned to form a hopper, where an initial deposition of granular particles and water forms the debris flow. The solid-water mixture in the hopper occupies a packing volume of 0.045 m³, with dimensions measuring 0.5 m $\times$ 0.3 m $\times$ 0.3 m ($L_p \times W_p \times H_p$) and uniform particle sizes $d_s = 10$ mm. To vary the water content within different debris flows, adjustments were made by modifying the initial solid volume fraction v_s in the hopper accordingly.

The initial solid volume fraction is defined the ratio of the solid volume to the total volume of the mixture packing. On the slope, situated at a distance of 2.5 m from the release gate, a slit dam measuring $B = 0.3$ m in width and $H = 0.5$ m in height is installed. The relative slit size s/d_s and transverse blockage $(B-s)/B$ obviously influences the peak impact force experienced by the slit dam and the regulation function of slit dam (Hu et al., 2020; Kong and Guan, 2023; Li et al., 2023; Zhou et al., 2019). This study adopts representative values of these two dimensionless size factors, $s/d_s = 6$, and $(B-s)/B = 0.8$, which are selected based on the typical ranges documented in existing literature (Choi et al., 2016; Gong et al., 2021; Goodwin and Choi, 2020; Kong et al., 2022; Marchelli et al., 2020; Zhou et al., 2020).

Regarding the boundary conditions, in the DEM simulation, we assumed that the right, left, and bottom faces formed a frictional wall boundary condition for the particles. In the CFD simulation domain, the upper face represented an atmosphere while the other five faces consisted of no-slip boundary walls. Initially, only fluid cells within the cubic debris domain are considered to contain water, whereas air occupied the remaining CFD fluid cells. Table 3 provides an overview of both model parameters and contact parameters for CFD-DEM. The time steps in DEM and CFD are 10⁻⁶ s and 10⁻⁴ s, respectively, such that coupling is made every 100 DEM cycles.

Prior to each simulation, the complete debris flow system is sub-

Table 3

Input parameters of the debris flow impact simulations.

DEM parameter	Value	CFD parameter	Value
Particle diameter d_s (mm)	10	Water density ρ_f (kg/m ³)	1000
Particle density ρ_s (kg/m ³)	2550	Water viscosity μ_f (Pa·s)	0.001
Particle Young's modulus E_s (GPa)	60	Air density ρ_a (kg/m ³)	1
Particle Poisson's ratio ν_s	0.25	Air viscosity μ_a (Pa·s)	1.48×10^{-5}
Coefficient of internal friction μ_s	0.48	Simulation parameters	Value
Coefficient of restitution e	0.78	Gravity g (m/s ²)	9.81
Barrier Young's modulus E_b (GPa)	3.2	DEM time step size Δt_s (s)	10 ⁻⁶
Barrier Poisson's ratio ν_b	0.35	CFD time step size Δt_f (s)	10 ⁻⁴

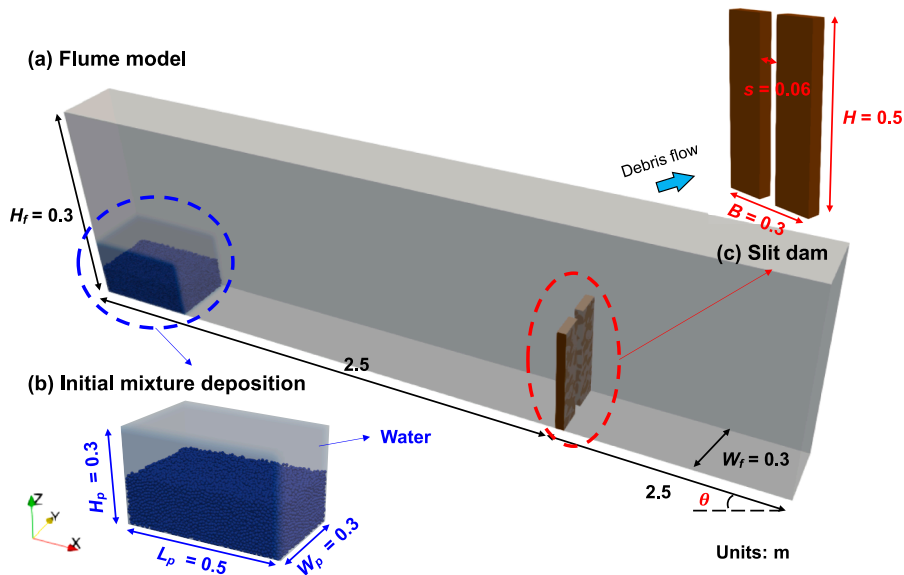
**Fig. 3.** (a) Model flume setup, (b) initial mixture deposition, and (c) slit dam model.

Table 4

Test plans and results of the interactions between the debris flows and the slit dam.

Test ID	$\theta(^{\circ})$	$\nu_s(-)$	$\rho(\text{kg/m}^3)$	$h_{f0}(\text{m})$	$v_{f0}(\text{m/s})$	$Fr(-)$
VS0.0-C10	10	0	1000	0.025	2.154	4.383
VS0.1-C10	10	0.1	1155	0.024	2.121	4.446
VS0.2-C10	10	0.2	1310	0.037	2.055	3.425
VS0.3-C10	10	0.3	1465	0.033	2.366	4.214
VS0.4-C10	10	0.4	1620	0.034	2.137	3.735
VS0.5-C10	10	0.5	1775	0.040	2.311	3.726
VS0.0-C20	20	0	1000	0.039	3.127	5.215
VS0.1-C20	20	0.1	1155	0.037	2.826	4.858
VS0.2-C20	20	0.2	1310	0.041	3.539	5.769
VS0.3-C20	20	0.3	1465	0.046	3.164	4.834
VS0.4-C20	20	0.4	1620	0.049	3.281	4.882
VS0.5-C20	20	0.5	1775	0.054	3.480	4.910
VS0.0-C30	30	0	1000	0.046	3.642	5.826
VS0.1-C30	30	0.1	1155	0.043	3.572	5.879
VS0.2-C30	30	0.2	1310	0.051	4.188	6.361
VS0.3-C30	30	0.3	1465	0.056	4.128	6.010
VS0.4-C30	30	0.4	1620	0.058	4.001	6.059
VS0.5-C30	30	0.5	1775	0.068	3.898	5.453
VS0.0-C40	40	0	1000	0.051	4.758	7.686
VS0.1-C40	40	0.1	1155	0.054	5.155	7.160
VS0.2-C40	40	0.2	1310	0.053	4.239	6.717
VS0.3-C40	40	0.3	1465	0.064	4.917	7.045
VS0.4-C40	40	0.4	1620	0.070	4.474	6.515
VS0.5-C40	40	0.5	1775	0.077	4.795	6.303

jected to a gravity of 9.81 m/s^2 . The particles are then randomly generated and dumped into the hopper. Subsequently, water is added to the hopper to form different debris flow systems with different initial soil volume fractions. Following the first deposition of the debris flows, the acceleration of gravity is changed to form the equivalent of different channel inclination angles, the debris flows are then released by

removing the release gate, and then the debris flows slide down to impact the slit dam.

2.4. Test plan

To test the effect of the channel inclination angle on the interaction behaviors between debris flows and slit dam, flume models with four different channel inclination angles were prepared: $\theta = 10^{\circ}$, 20° , 30° and 40° , respectively. For each channel inclination angle, the fluid properties (water volume fractions) varied between 1 (pure water) and 0.5 to investigate the effect of the water content on the flow–dam interaction. Six values of water volume fractions ν_w were adopted, namely, 1, 0.9, 0.8, 0.7, 0.6, and 0.5, corresponding to solid volume fractions ν_s of 0, 0.1, 0.2, 0.3, 0.4, and 0.5, respectively, covering the range from water flows to hyper-concentrated flows (Pierson, 2005). A total of 24 cases were simulated. The test plans and related simulation results of the interactions between the debris flows and the slit dam are summarized in Table 4. Fr represents the Froude number, a widely used parameter for quantifying the ratio of inertial and gravitational forces:

$$Fr = \frac{v_{f0}}{\sqrt{gh_{f0}\cos\theta}} \quad (12)$$

where v_{f0} and h_{f0} denote the frontal flow velocity, and maximum flow depth, respectively, because the accurate determination of frontal flow velocity and maximum flow depth within the foremost region is essential for accurately characterizing the run-up height and impact load (Ng et al., 2019). As shown in Fig. 4, these values are computed when the debris flow passes through a static measuring volume positioned upstream from the slit dam under the open-channel flows condition. To ensure no airborne particles are overlooked, this measuring volume spans both the entire width (0.3 m) of the flume and height (1.0 m) of sidewalls. The thickness of this volume is set at 50 mm to be five times

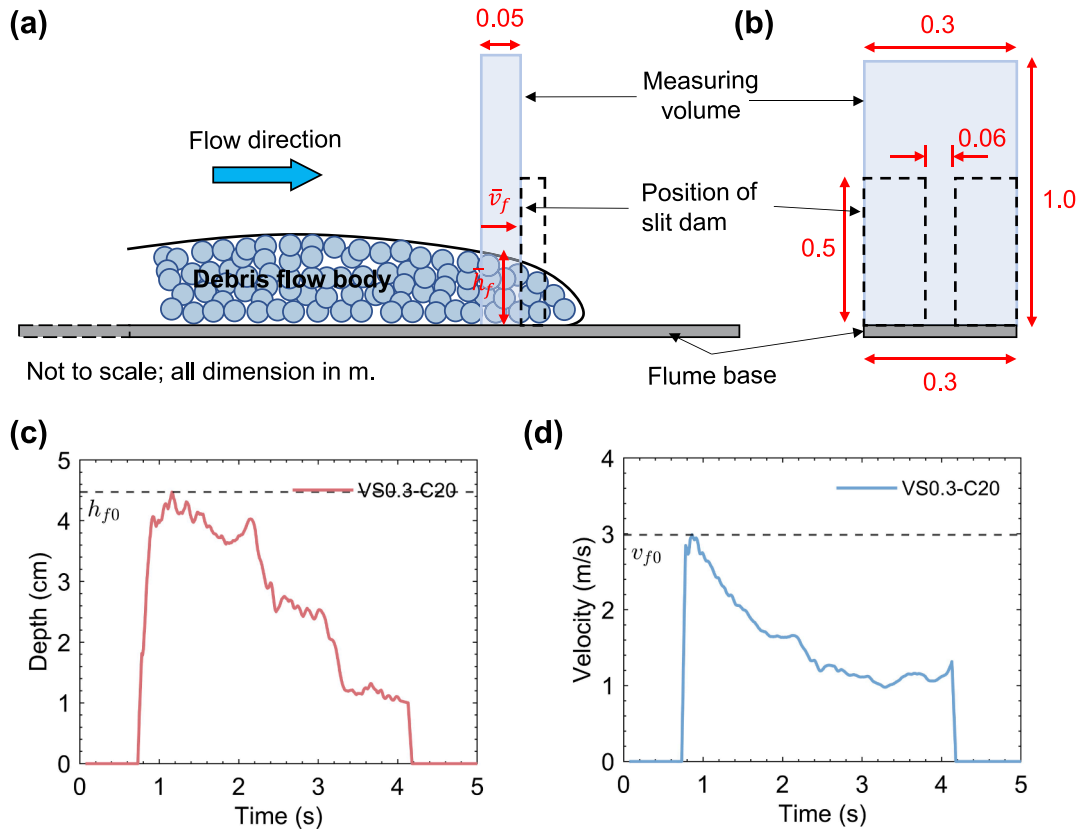


Fig. 4. (a) Side-view, and (b) front-view schematic of the flume model and measuring volume without slit dam setup; and characteristic curves for open-channel flows: (c) flow depth, (d) frontal flow velocity. The dashed lines in (c) and (d) represent the maximum approaching flow depth and velocity.

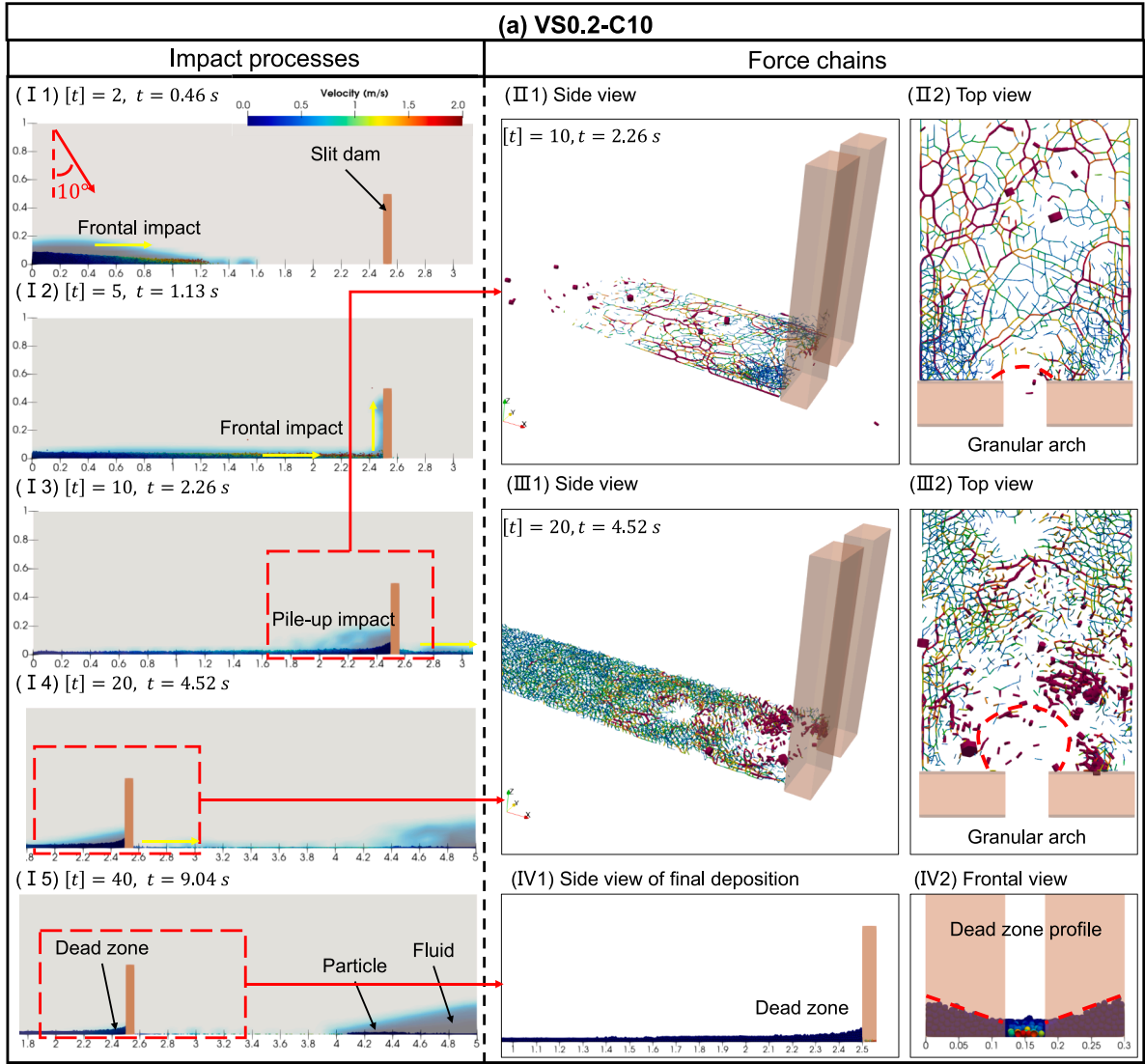


Fig. 5. Impact processes of debris flows with (a) low ($\nu_s = 0.2$) and (b) high ($\nu_s = 0.5$) solid volume fraction against a slit dam with a channel inclination angle of $\theta = 10^\circ$ (gentle slope, $\theta/\varphi < 1$): (I) Typical snapshots of impact progress of debris flow at $[t] = t/\sqrt{H/g} = 2, 5, 10, 20$ and 40 ; (II-III) side views and (c) top views of force chains of particles within debris flow at $[t] = 10, 20$; (IV) side view and frontal view of final deposition at $[t] = 40$.

larger than particle sizes ($d_s = 10\text{mm}$) employed in simulations. As for the water phase, we continuously monitor real-time variations in interface heights ($\alpha_f = 0.5$) and average velocity throughout the measuring volume in OpenFOAM.

3. Numerical simulation results

The following analyses utilize dimensionless variables to enable generalization of research outcomes. In particular, the normalized simulation time is $[t] = t/\sqrt{H/g}$, where H represents the height of slit dam. And the impact force F is normalized by mg , where m is the initial mass of the debris flow two-phase system.

3.1. Impact kinematics

To investigate the dynamic process of flow–slit dam interactions and assess the influences of channel inclination angle and water content, this section analyzes the propagation and interaction processes. Fig. 5 illustrates the impact kinematic process of debris flows against a slit dam with low ($\nu_s = 0.2$) and high ($\nu_s = 0.5$) solid volume fractions at five

typical moments, along with corresponding contact force chain networks when the channel inclination angles is $\theta = 10^\circ$ (gentle slope, $\theta/\varphi < 1$, where $\varphi = 26.2^\circ$ determined by angle of repose test). In each panel, the granular material and the water motion profile are marked as spheres and blue background, respectively. The thickness and color of tubes serve as indicators for the magnitude of inter-particle contact force. With a low solid volume fraction ($\nu_s = 0.2$), shown in Fig. 5(a), after the debris flows were released, the initial debris deposition collapses and flows with a low velocity along the flume, forming a long and thin debris body. As the debris flow front reaches the slit dam, a few particles either pass through the mouth or bounce back from barriers on both sides; however, the majority of the particles are deposited on the flume base before the slit dam, forming a dead zone (a volume of stationary material develops upslope from the barrier) with a slender tail. Note that Kong et al., (2021b) proposed a modified granular temperature that can calculate the region of impact-induced dead zones during debris flows against a close-type rigid dam. Additionally, the presence of strong contact force chains within subsequent flows highlights the pivotal role of the dead zone in providing a crucial buffering effect. Furthermore, inconspicuous granular arches are observed in close

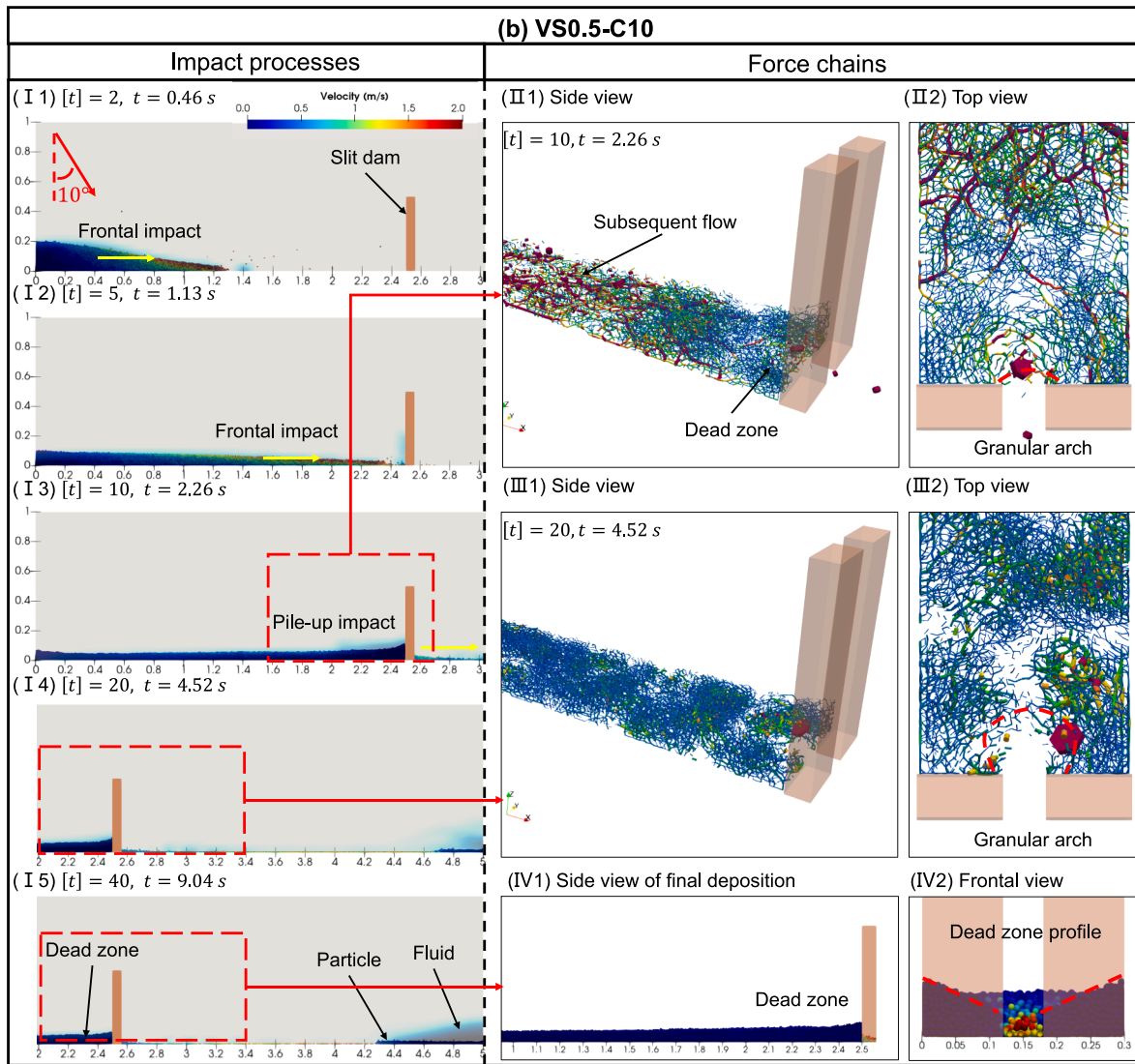


Fig. 5. (continued).

proximity to the slit. With a high solid volume fraction ($v_s = 0.5$), as depicted in Fig. 5(b), it can be observed that there is no obvious disparity in the overall flow impact and interaction process when compared to VS0.2-C10. The majority of solid particles are intercepted by the flume base friction, resulting in the formation of an elongated and slender dead zone before slit dam, while the liquid phase gradually passes through the slit downstream after an upward jet.

To make a comparison, Fig. 6 illustrates the interaction processes of the debris flows with low ($v_s = 0.2$) and high ($v_s = 0.5$) solid volume fractions against the slit dam at five typical moments when the channel inclination angle is $\theta = 30^\circ$ (steep slope, $\theta/\varphi > 1$). For a low solid volume fraction ($v_s = 0.2$), shown in Fig. 6(a), the debris flow forms a thick debris body after the debris flow system is released. The debris flow front can then arrive at the slit dam at high speeds in less time. Then, the debris flow sharply ascends the slit dam and shows an obvious overtopping flow phenomenon. The subsequent high-velocity flow can continue to climb the material deposited in the dead zone and pile up, ultimately forming a dead zone with a shorter tail. For a high solid volume fraction ($v_s = 0.5$), shown in Fig. 6(b), a more obvious overtopping flow phenomenon is observed and a larger ramp-like dead zone whose height is reduced from the sidewall to slit is formed. Additionally, obvious granular arches are observed near the slit.

By analyzing Figs. 5 and 6, it is evident that both the channel

inclination angle and water content (solid volume fraction) exert significant influence on the impact kinematics of flow–slit dam interactions. The deposition of debris flows on gentle slopes ($\theta/\varphi < 1$) is primarily influenced by flume basal friction, irrespective of the presence of a slit dam, highlighting the crucial role played by energy dissipation and deceleration. As for steep slopes ($\theta/\varphi > 1$), the solid volume fraction plays an important role in the formation and destruction of stable granular arches, thus affecting the flow–slit dam interaction.

3.2. Slit dam response

The engineering design of a slit dam capable of withstanding debris flows require the estimation of impact forces (Zhang and Huang, 2022). Therefore, it is crucial to comprehend the influence of solid volume fraction and channel inclination angle on the impact forces exerted on the slit dam. The evolution of the components and total impact force for debris flows with various initial solid volume fractions at $\theta = 20^\circ$ is illustrated in Fig. 7. F_{total} is the total impact force, F_s is the solid-resulted impact force, and F_f is the fluid-resulted impact force. The values of F_f and F_s are derived from the pressures exerted by the boundary cells in CFD and the particle–wall interaction forces in DEM, respectively. The results demonstrate that an increase in the initial solid volume fraction leads to a noticeable increasing in the total impact force, while

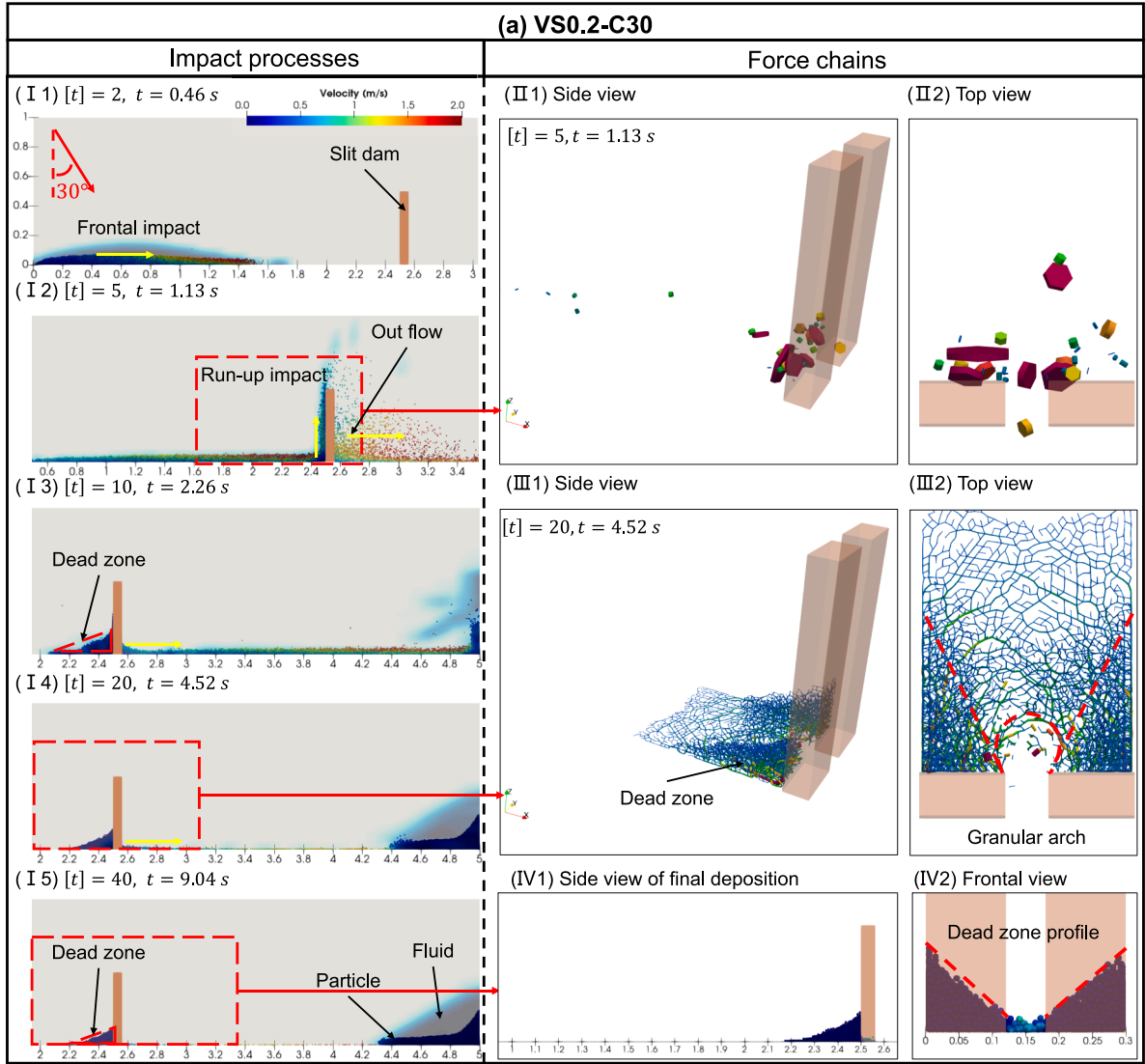


Fig. 6. Impact processes of debris flows with (a) low ($\nu_s = 0.2$) and (b) high ($\nu_s = 0.5$) solid volume fraction against a slit dam with a channel inclination angle of $\theta = 30^\circ$ (steep slope, $\theta/\varphi > 1$): (I) Typical snapshots of impact progress of debris flow at $[t] = t/\sqrt{H/g} = 2, 5, 10, 20$ and 40 ; (II–III) side views and (c) top views of force chains of particles within debris flow at $[t] = 10, 20$; (IV) side view and frontal view of final deposition at $[t] = 40$.

simultaneously highlighting the increasingly prominent multimodal fluctuation characteristics. Additionally, it can be inferred that, for a given initial total volume, an increase in the initial solid volume fraction will lead to a gradual increase in F_s and its proportion within F_{total} . For example, the proportions of F_s within F_{total} increase from 15.35 % of VS0.1-C20 to 39.04 % of VS0.5-C20. Meanwhile, F_f will exhibit minimal changes but with an increasing level of fluctuation, attributed to particle–fluid interaction.

Figures 8 and 9 further compare the evolution of the total impact force for debris flows with different initial solid volume fraction and channel inclination angle, respectively. Fig. 10 plots the normalized peak impact forces against different solid volume fraction and channel inclination angle. Fig. 8 implies that the total force exerted by debris flows over time is significantly affected by the initial solid volume fraction. In general, the fluid phase in debris flows significantly increases its mobility and therefore its dynamic impact forces. However, as shown in Fig. 10, when solid volume fraction is more than 0.3, the normalized total impact force F_{total}/mg exhibited no significant variation, remaining essentially unaltered. The formation of a larger dead zone due to the increase in initial solid volume fraction is likely

responsible for this phenomenon, as the buffering effect of the dead zone plays a crucial role in attenuating the impact force. Moreover, for gentle slope (e.g., 10° , shown in Fig. 9(a)), the normalized total impact force F_{total}/mg with different initial solid volume fraction have little difference (Fig. 10, $\tan\theta = 0.18$). The possible explanation could be that a majority of debris deposition adhere to the surface of the flume base due to particle–flume friction, thereby posing challenges for subsequent flow towards the slit dam. This presents a challenge in terms of increasing the impact force. For steep slopes (e.g., $30^\circ, 40^\circ$, shown in Fig. 9(c, d)), F_{total}/mg increases with the increase of solid volume fraction (Fig. 10, $\tan\theta = 0.58, 0.84$). The possible explanation could be that the steep slope facilitates the debris flow to overcome base surface friction, thereby promoting sufficient debris flow for impacting the slit dam, causing subsequent and overtopping processes, ultimately resulting in an increase in impact force.

3.3. Run-up height

Debris flows impact slit dams, and run up along barriers, having the potential to overflow them (Ng et al., 2016), just like shown in Fig. 6.

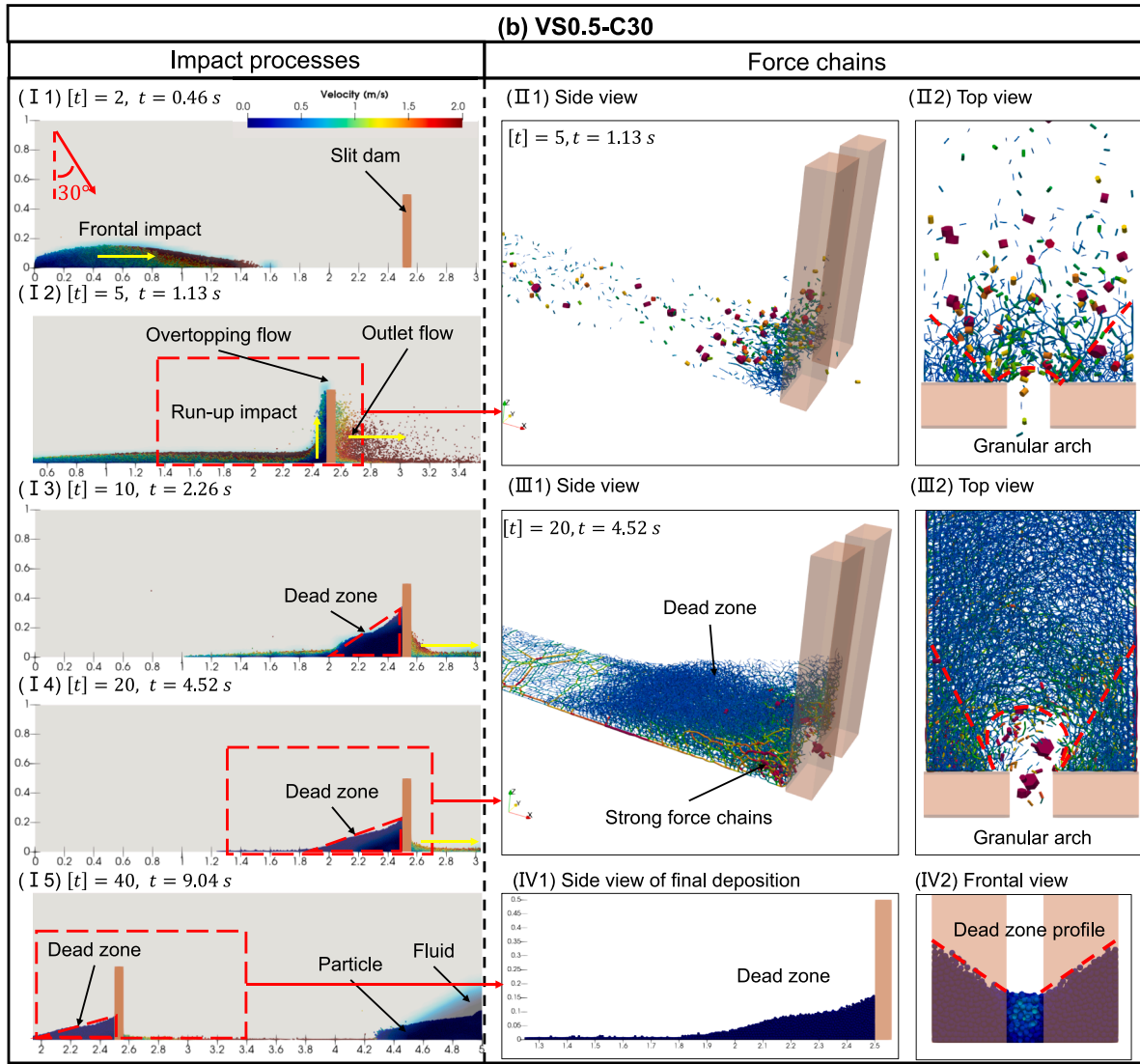


Fig. 6. (continued).

Accordingly, the run-up height is one of the most important indicators for slit dam design (Choi et al., 2016; Kwan et al., 2015). However, current research has primarily focused on dry granular flows (Zhou et al., 2020) and may not be able to accurately estimate the run-up heights of actual debris flows in nature (Iverson et al., 2016). In addition, the interaction process between a debris flow and a slit dam is fairly complicated, with the inclusion of downstream discharge, unlike in the case of closed-type dams, adding to the complexity of the problem. In this study, pure water was utilized as a surrogate for the liquid phase in debris flow. However, it is important to note that this approach may result in particle segregation from the liquid, particularly at higher velocities (Fang et al., 2022). Therefore, instead of directly measuring the run-up heights of debris flows, we have opted to assess the run-up heights of particles using Eq. (13):

$$h_f = \frac{2}{N} \sum_{i=1}^N h_i \quad (13)$$

where N is number of particles in measuring volume positioned upstream from the slit dam at any moment (see Fig. 4, and Fig. 11(f)); h_i is the height coordinate of the i^{th} particle.

Fig. 11 displays the time evolution series of the run-up height for debris flows with various initial solid volume fraction. The run-up height h_f is normalized by the height of the slit dam H . Numerical simulation

results reveal that the run-up height in each case rapidly reaches its peak values following a similar trend, subsequently undergoes a gradual decrease, reflecting the process of downstream discharge, and ultimately stabilizes at a constant value. Fig. 12 displays the time evolution series of the run-up height for debris flows with various channel inclination angle. The results indicate that, for gentle slope (e.g., 10° , shown in Fig. 12(a)), the normalized run-up heights gradually increase until they reach a maximum value, which is significantly lower compared to other angle conditions and does not exceed 0.4. Then, the run-up heights reach a steady state with a slight decrease, indicating that majority of the debris flows are intercepted by the friction of flume base. Additionally, a positive correlation between the run-up heights and the solid volume fraction is observed for v_s values below 0.3. However, no further increase in run-up heights is observed beyond v_s exceeding 0.3. These results indicate that run-up height is influenced by both the solid volume fraction and channel inclination angle. For steep slopes (e.g., 30° , shown in Fig. 12(c)), the run-up height continuously increases until reaching its peak value. Under the same inclination angle, different solid volume fractions yield approximately equal peak run-up heights. Subsequently, it descends along an almost coinciding curved trajectory before bifurcating. The start time of curve bifurcating and the duration of run-up heights decreasing, that is clogging time of granular arches, are evidently affected by the solid volume fractions.

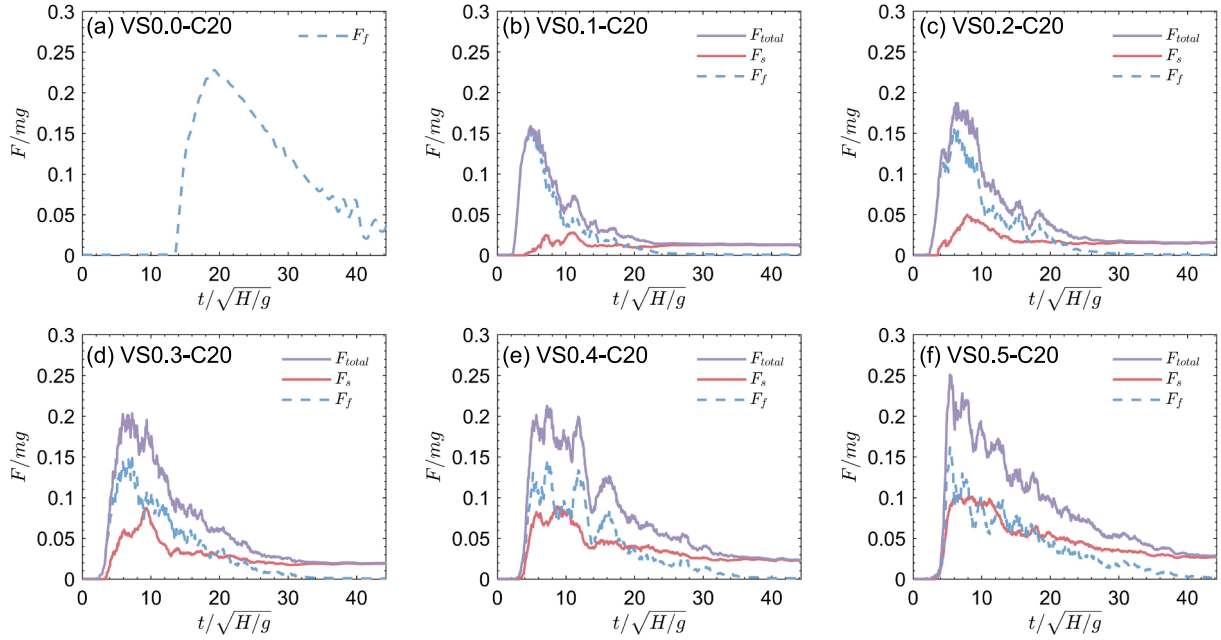


Fig. 7. Evolution of the impact force for debris flows with various initial solid volume fraction: (a) VS0.0-C20, (b) VS0.1-C20, (c) VS0.2-C20, (d) VS0.3-C20, (e) VS0.4-C20, and (f) VS0.5-C20. Note, F_{total} is the total impact force, F_s is the solid-ruled impact force, and F_f is the fluid-ruled impact force.

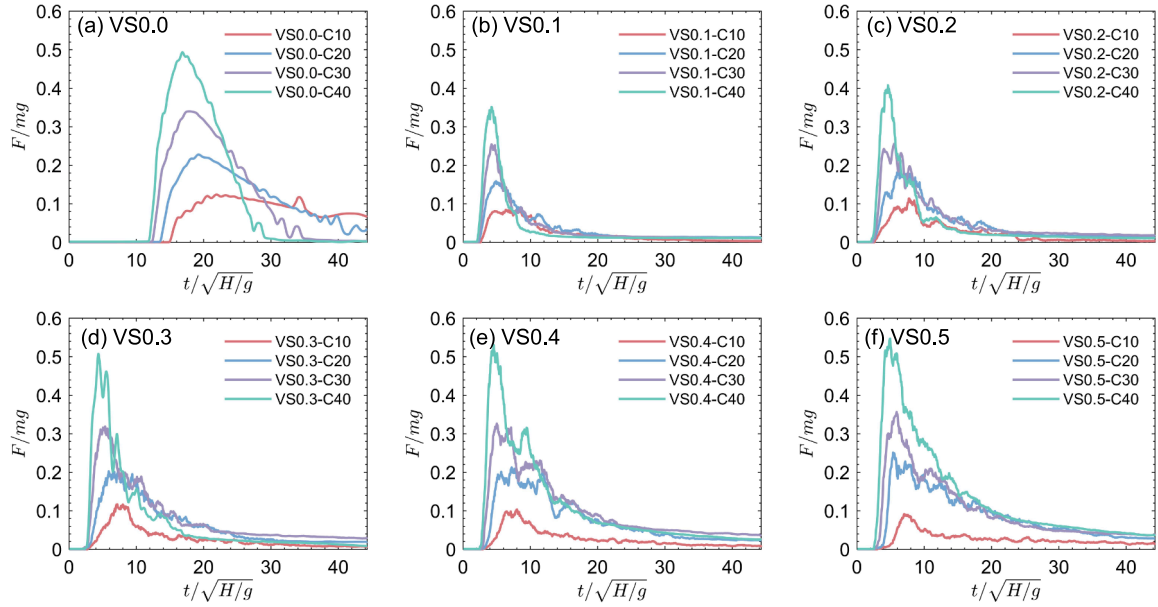


Fig. 8. Evolution of the total impact force for debris flows with various initial solid volume fraction: (a) $v_s = 0.0$, (b) $v_s = 0.1$, (c) $v_s = 0.2$, (d) $v_s = 0.3$, (e) $v_s = 0.4$, and (f) $v_s = 0.5$.

To further understand how water content and inclination angle affects the peak run-up height, we have plotted the normalized peak run-up heights against different initial solid volume fraction and channel inclination angle in Fig. 13. As shown in Fig. 13(a), when flume slope is gentle (e.g., 10°), there is a critical solid volume fraction value, $v_s = 0.3$ in this study, below which the solid volume fraction assumes significance, leading to a predominant distribution of particles at the flume base before the slit dam, primarily governed by “trapping” mechanical pattern proposed by Goodwin and Choi, (2020). Conversely, when exceeding this critical value, due to the low velocity of subsequent flow, reaching the dead zone becomes challenging and consequently has minimal impact on run-up height. However, the run-up height for steep slope (e.g., 30° – 40°) cases is minimally influenced by the solid volume

fraction, but exhibits an increase with the flume inclination angle (Fig. 13(b)). The possible reasons can be identified as follows: (i) The range of Froude number (Fr) $\theta = 30^\circ, 40^\circ$ in this study is relatively limited (5.8–7.2), its impact on the run-up height might be not statistically significant. (ii) The wider slit $s/d_s = 6$ poses challenges in achieving the formation of a stable granular arch during early high-speed impact stage. After the formation of dead zone, the velocity of subsequent flow is so high that it can continue to climb over the dead zone, and pass through the slit because of the contraction effect, causing difficulty in further accumulation (Rossi and Armanini, 2019; Huang et al., 2023).

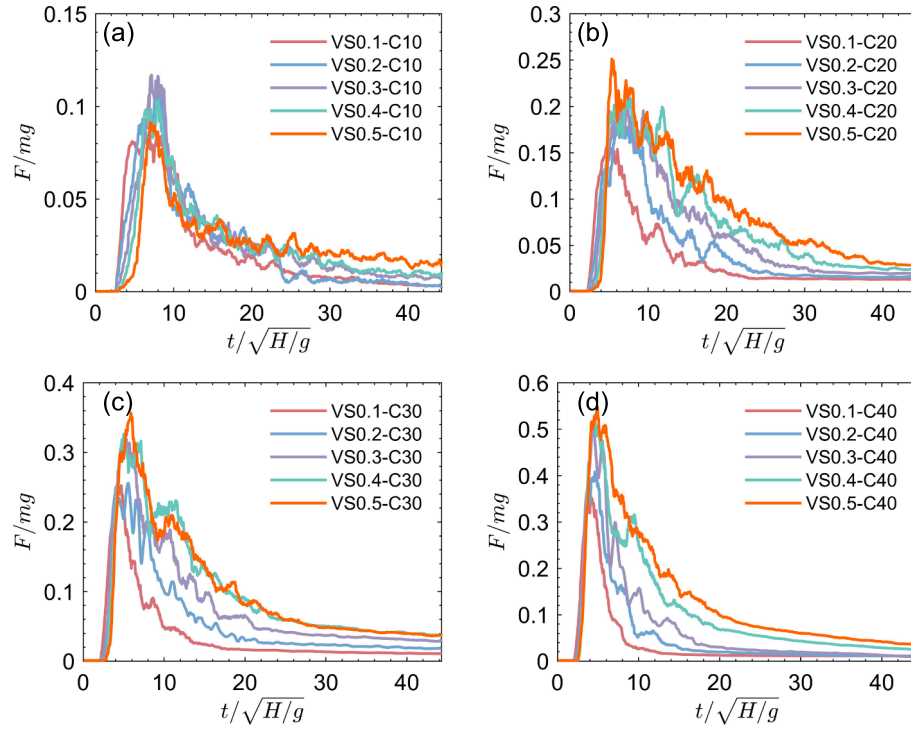


Fig. 9. Evolution of the total impact force for debris flows with various channel inclination angle: (a) $\theta = 10^\circ$, (b) $\theta = 20^\circ$, (c) $\theta = 30^\circ$, and (d) $\theta = 40^\circ$.

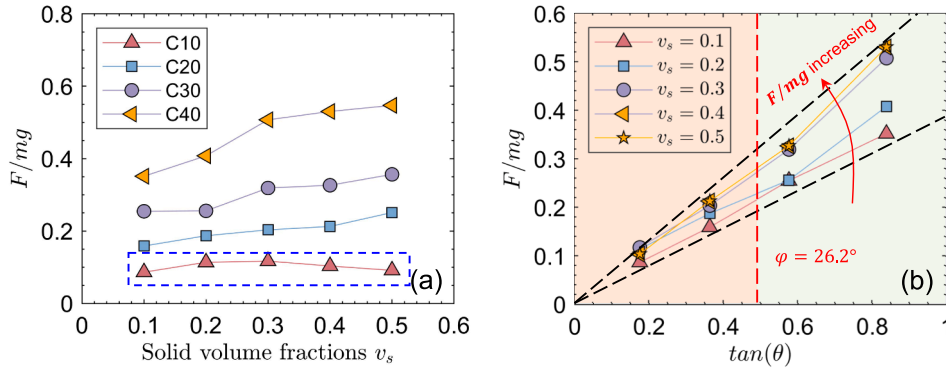


Fig. 10. Effect of the (a) initial solid volume fraction and (b) channel inclination angle on the peak total impact force on slit dam.

3.4. Retention efficiency

One of the functions of the slit dam is to trap large particle in debris flows, therefore reducing the risk of destruction to downstream infrastructure. Although a high retention efficiency is desirable, it can lead to structures being filled too easily and losing their intended regulatory function. Therefore, deciding on a reasonable retention efficiency for slit dam designs is imperative. The retention efficiency in this study is defined as the ratio of particles behind the slit dam to the initial number of particles without regard to fluid phase.

The evolution of the retention efficiency with various initial solid volume fraction and channel inclination angle are presented in Figs. 14 and 15, respectively. As shown in Figs. 14 and 15, numerical simulation results reveal that the observed evolutionary trend consistently follows a two-stage pattern across all temporal curves: in the early stage of interaction between the debris flow and the slit dam, the retention efficiency is abruptly decreases due to the occurrence of overtopping flow phenomenon and the subsequent high-velocity passage through the slit. Subsequently, the periodic formation and destruction of granular arches

will significantly impede the flow of debris from the narrow opening, resulting in a decelerated temporal curvature. In addition, it can be found that when the channel inclination angle is 10° , the vast majority of the debris flow are restrained by the friction of the flume base and are arrested by the slit dam. And the retention efficiency will rapidly decrease as the slope becomes steeper.

To further understand how water content and inclination angle affect the retention efficiency, we have plotted the retention efficiency against different mass water content which is converted from solid volume fraction for a convenient comparison with experimental results by Zhou et al., (2019) and Kong and Guan, (2023), and channel inclination angle in Fig. 16. As shown in Fig. 16(a), the retention efficiency decreases as the mass water content increases. It is worth noting that the two-phase flows simulated in this study encompass a wide range, spanning from water flow to hyper-concentrated flows, while the range of the water content was from 18 % to 38 % in the study of Zhou et al., (2019). And the findings illustrated in Fig. 16(b) demonstrate a decline tendency in retention efficiency as the channel inclination angle increases. As for 10° inclination angle, a majority of the debris flow are trapped by the

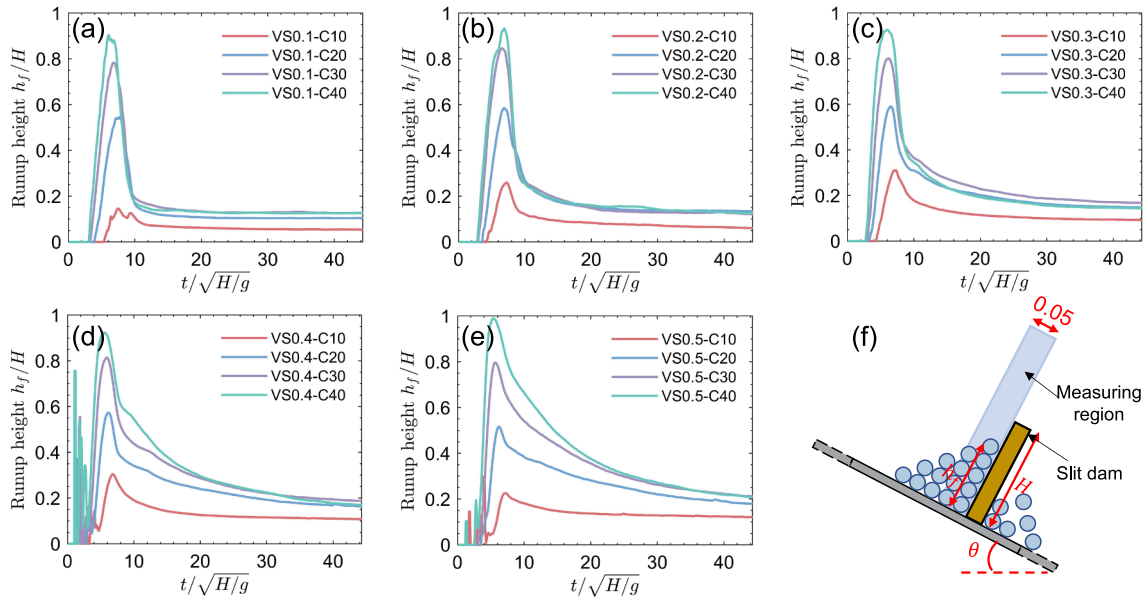


Fig. 11. Evolution of the run-up height for debris flows with various initial solid volume fraction: (a) $v_s = 0.1$, (b) $v_s = 0.2$, (c) $v_s = 0.3$, (d) $v_s = 0.4$, (e) $v_s = 0.5$.

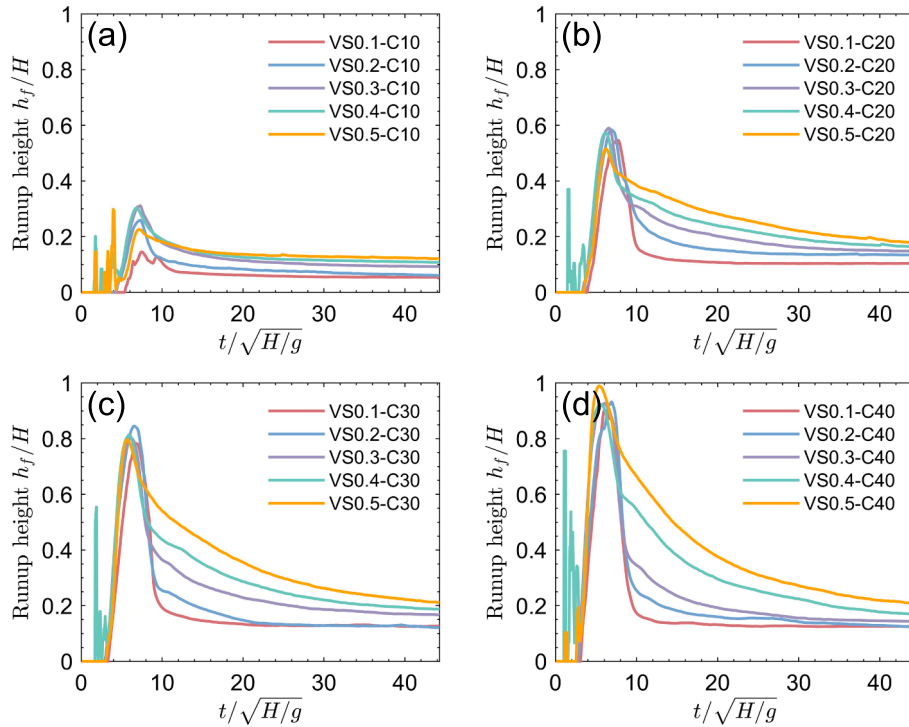


Fig. 12. Evolution of the run-up height for debris flows with various channel inclination angle: (a) $\theta = 10^\circ$, (b) $\theta = 20^\circ$, (c) $\theta = 30^\circ$, and (d) $\theta = 40^\circ$.

friction of flume base, leading to higher retention efficiencies. However, for 40° inclination angle, a “self-cleaning” pattern occurs because the granular arches do not work under high-velocity impact. Undoubtedly, the larger transverse blockage (0.8 in this study) also plays a crucial role. As for the angle from 10° to 40° , the periodic formation and destruction of granular arches will significantly influence the retention efficiency.

4. Discussion

Kong et al. (2021a) treated the fluid phase in a debris flow as a non-Newtonian fluid modeled with the Herschel-Bulkley model. In this study, pure water (Newtonian fluid) was employed to simulate the liquid

phase in debris flow. However, this approach may lead to particle separation from the liquid system, particularly at higher velocities (Fang et al., 2022), which is highly probable during the interaction between debris flows and slit dam. Furthermore, it should be noted that debris flows can encompass a wide range of particle sizes ranging from fine clay to boulders. Therefore, it is more appropriate to consider the liquid phase as a non-Newtonian fluid due to the presence of water and clay, which can significantly influence the density and effective viscosity of the fluid phase (Iverson, 1997).

Additionally, the existing methods of initiating depositions and release of granular flows can be classified into two categories: dam-break (Huang et al., 2022; Marchelli et al., 2020; Zhang et al., 2021),

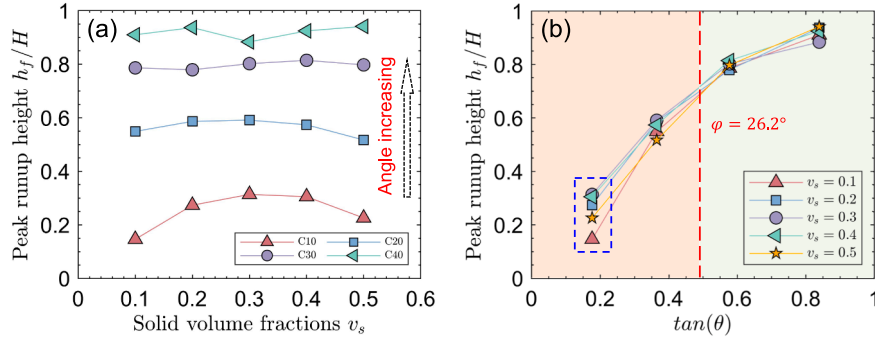


Fig. 13. Effect of the (a) initial solid volume fraction and (b) channel inclination angle on the peak run-up height.

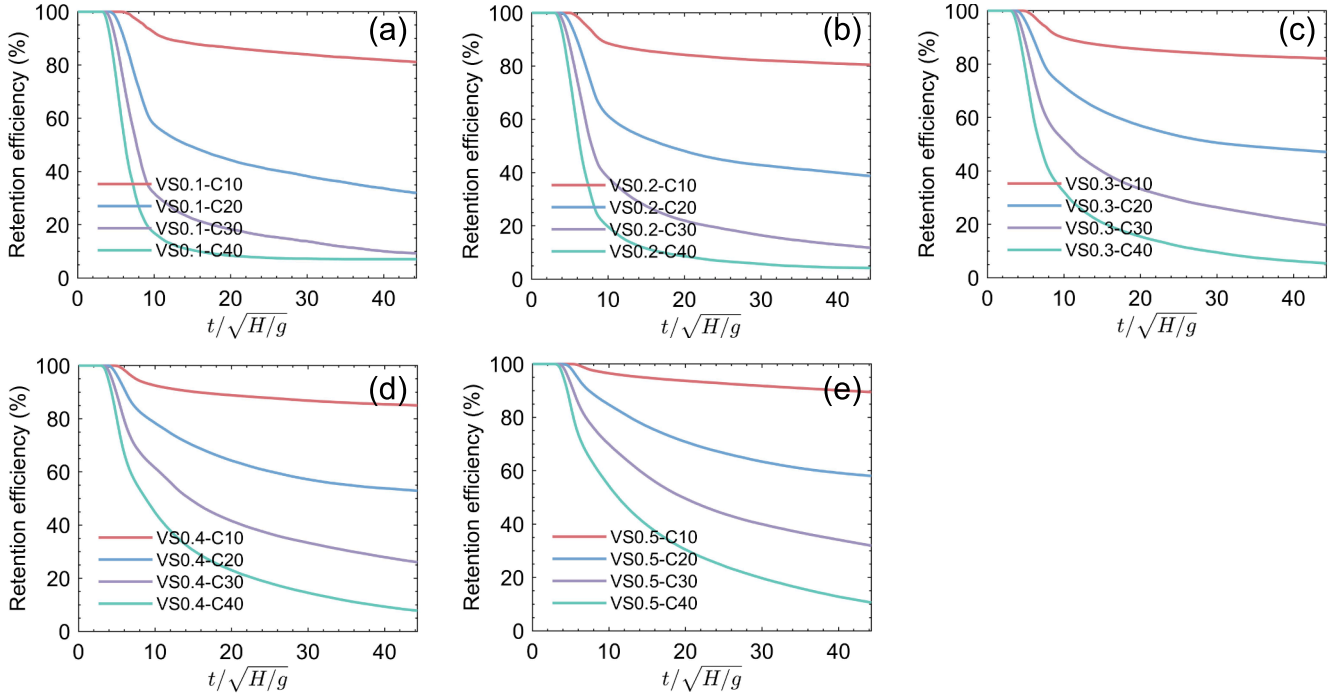


Fig. 14. Evolution of the retention efficiency for debris flows with various initial solid volume fraction: (a) $v_s = 0.1$, (b) $v_s = 0.2$, (c) $v_s = 0.3$, (d) $v_s = 0.4$, (e) $v_s = 0.5$.

and continuous flux (Kong and Guan, 2023; Zhou et al., 2020). The former can better align with the results of the flume test, thereby verifying their effectiveness, while also offering a shorter simulation time and reduced computational costs. On the other hand, the latter excels in reproducing the long-term characteristics of interactions between natural debris flow and slit dams (Leonardi et al., 2019). Therefore, it is imperative to further compare and validate both approaches in terms of their rationality and influence on the interaction process in future studies.

5. Conclusions

This paper presented a multi-scale simulation study of debris flows impacting a slit dam to examine the influences of the initial solid volume fraction and channel inclination angle on interactions between the debris flows and the slit dam, as well as the regulation function of the slit dam by CFD-DEM. On the basis of the limited numerical simulation configuration and our analysis, the conclusions of this study can be summarized as follows.

1. The propagation and interaction processes of debris flows with varying initial solid volume fraction in different channel inclination configurations against a slit dam were captured and systematically analyzed. The results indicate that both the initial solid volume fraction and the channel inclination angle have significant effects on the impact kinematics of flow-slit dam interactions. For gentle slopes ($\theta/\varphi < 1$), the deposition of debris flows is primarily influenced by flume basal friction, whereas for steep slopes ($\theta/\varphi > 1$), an obvious overtopping flow phenomenon can occur, ultimately forming a dead zone with a shorter tail because of the formation and destruction of stable granular arches.
2. With respect to the response of the rigid slit dam, an increase in the initial solid volume fraction leads to a noticeable increasing in the total impact force through increasing the magnitude and proportion in solid-resulted impact force F_s , however, fluid-resulted impact force F_f will exhibit minimal changes but with an increasing level of fluctuation.
3. The evolution and peak value of impact force are governed by the combined influence of the initial solid volume fraction and the channel inclination angle. For gentle slopes ($\theta/\varphi < 1$), the normalized total impact forces with different initial solid volume fraction

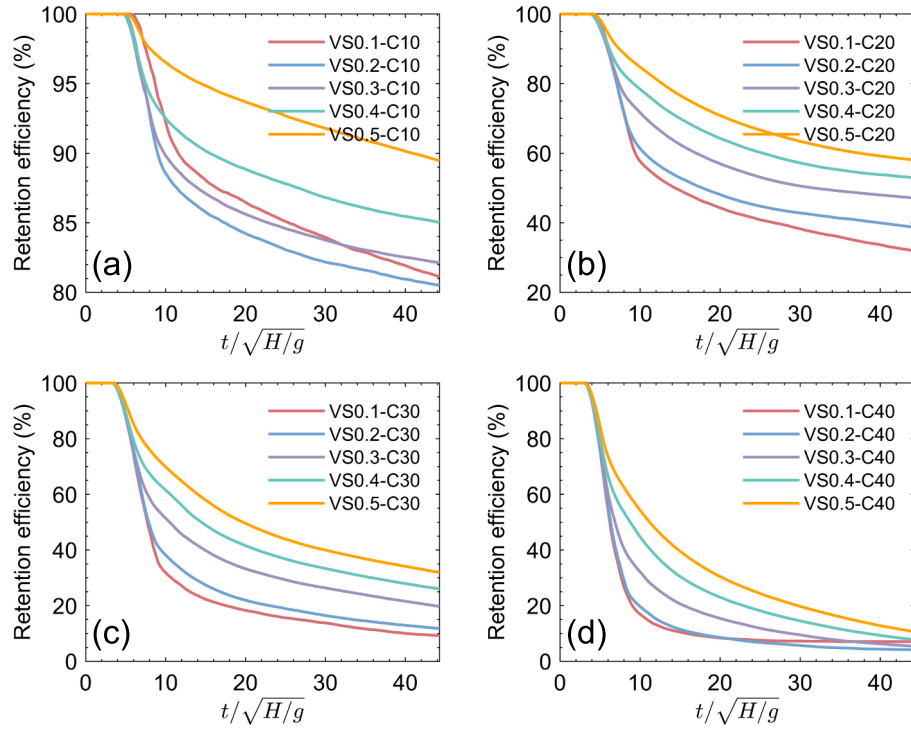


Fig. 15. Evolution of the retention efficiency for debris flows with various channel inclination angle: (a) $\theta = 10^\circ$, (b) $\theta = 20^\circ$, (c) $\theta = 30^\circ$, and (d) $\theta = 40^\circ$.

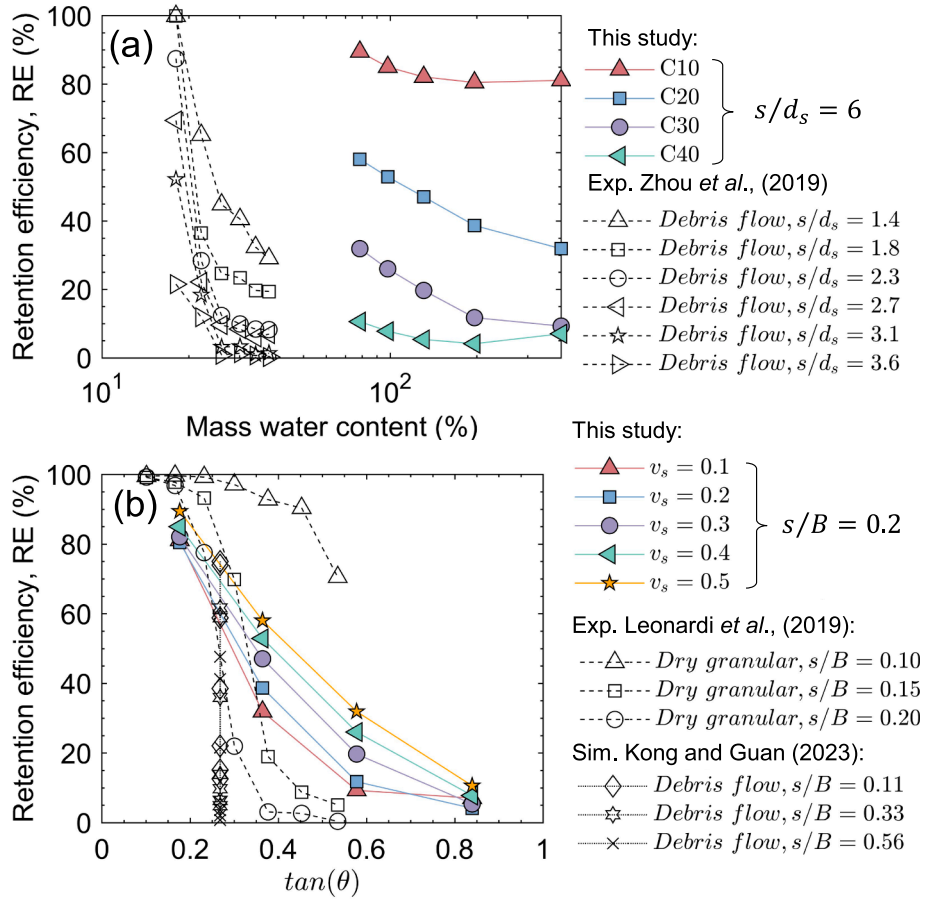


Fig. 16. Effect of the (a) mass water content and (b) channel inclination angle on the retention efficiency.

have little difference because of the flume basal friction presents a challenge for subsequent flow towards the slit dam. However, for steep slopes ($\theta/\varphi > 1$), the normalized total impact forces increase with the increase of solid volume fraction due to overcoming the base surface friction.

4. For gentle slopes, a positive correlation between the run-up heights and the solid volume fraction v_s is observed when $v_s < 0.3$. However, no further increase in run-up heights is observed when $v_s > 0.3$. For steep slopes, under the same inclination angle, different solid volume fractions yield approximately equal peak run-up heights.
5. The evolutionary trend of retention efficiency consistently follows a two-stage pattern, controlled by the periodic formation and destruction of granular arches near slit. And the retention efficiency will rapidly decrease as the slope becomes steeper.

CRediT authorship contribution statement

Yandong Bi: Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation, Conceptualization. **Yu Huang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition. **Bei Zhang:** Writing – original draft, Visualization, Software, Data curation. **Jian Pu:** Writing – review & editing, Validation, Software, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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